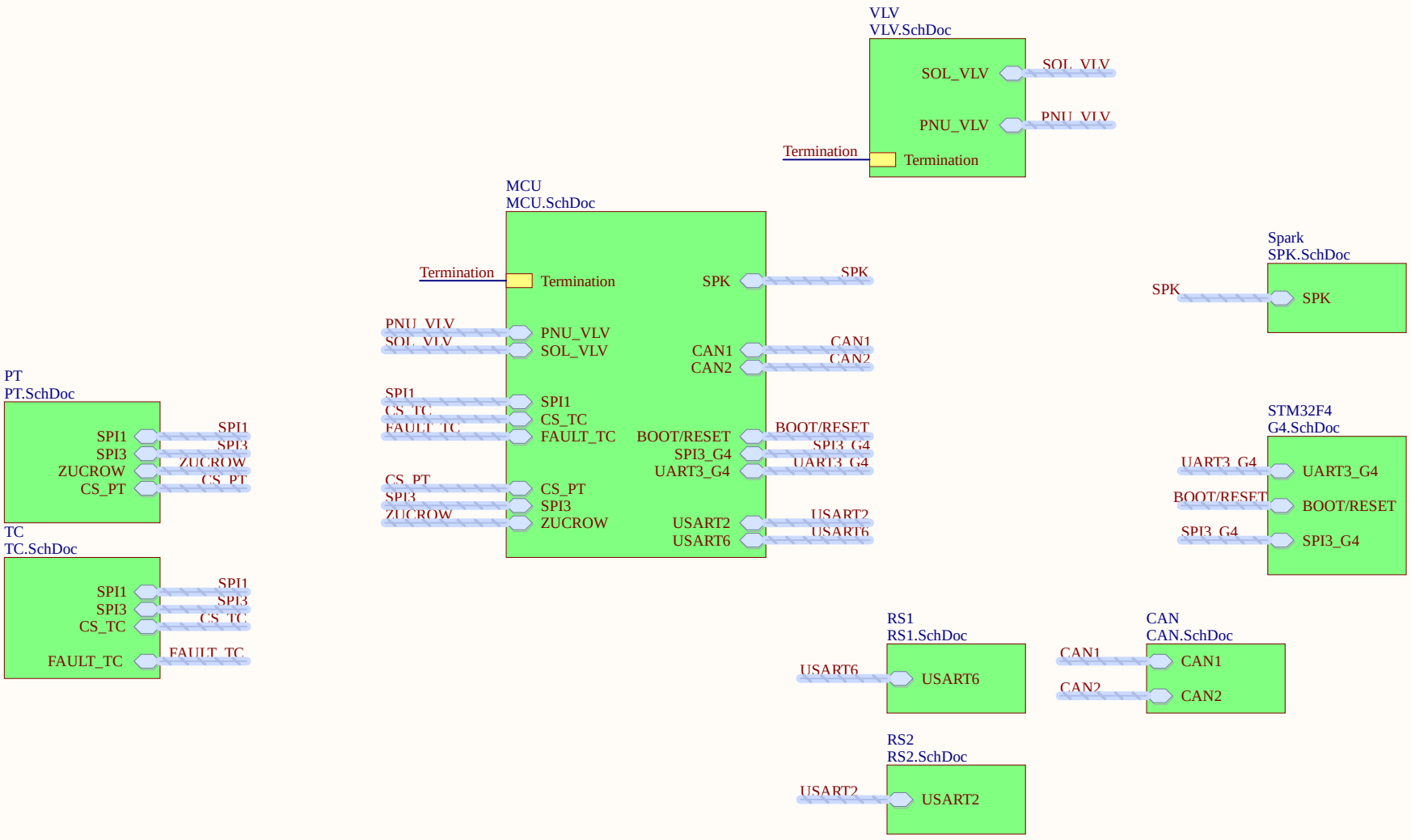
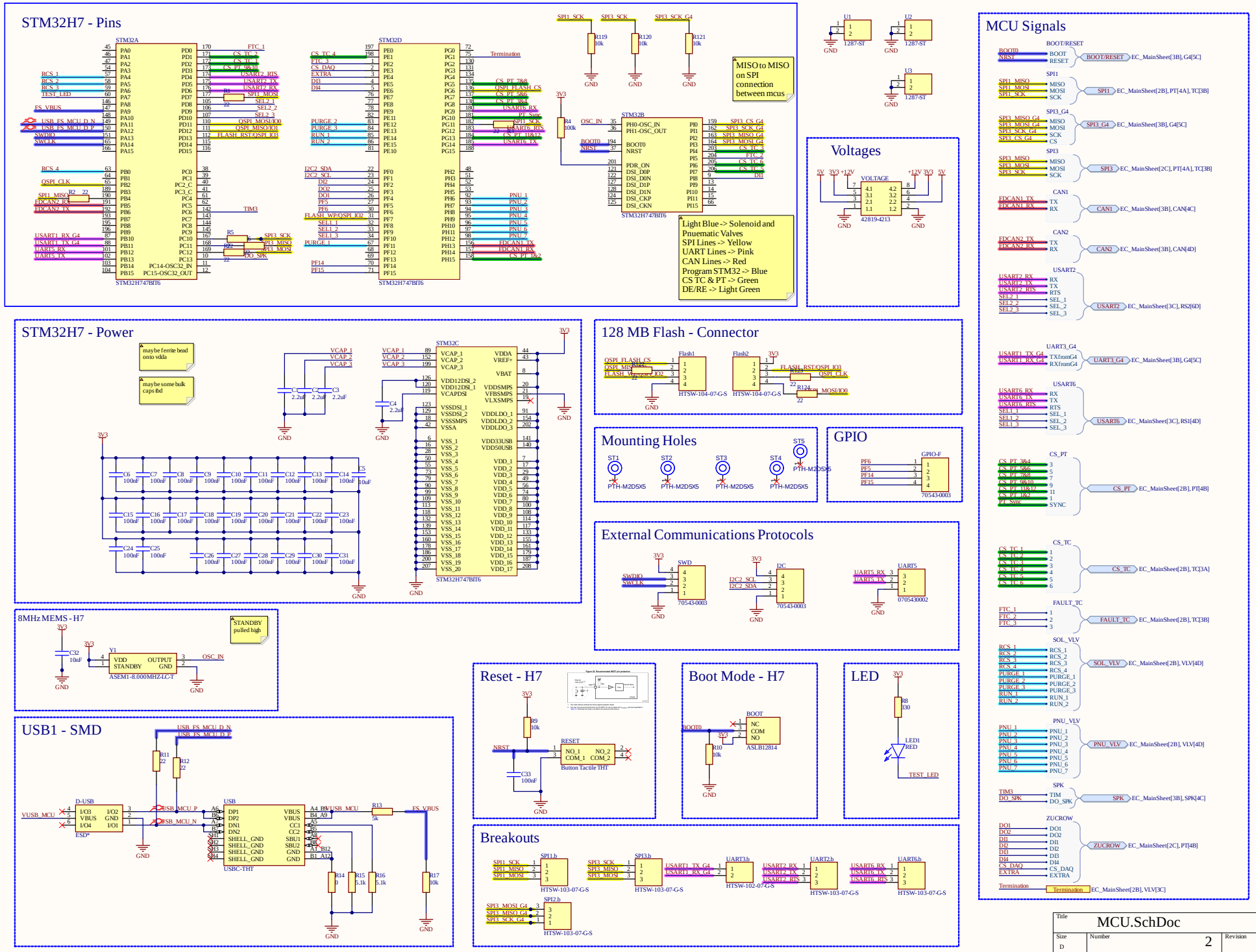
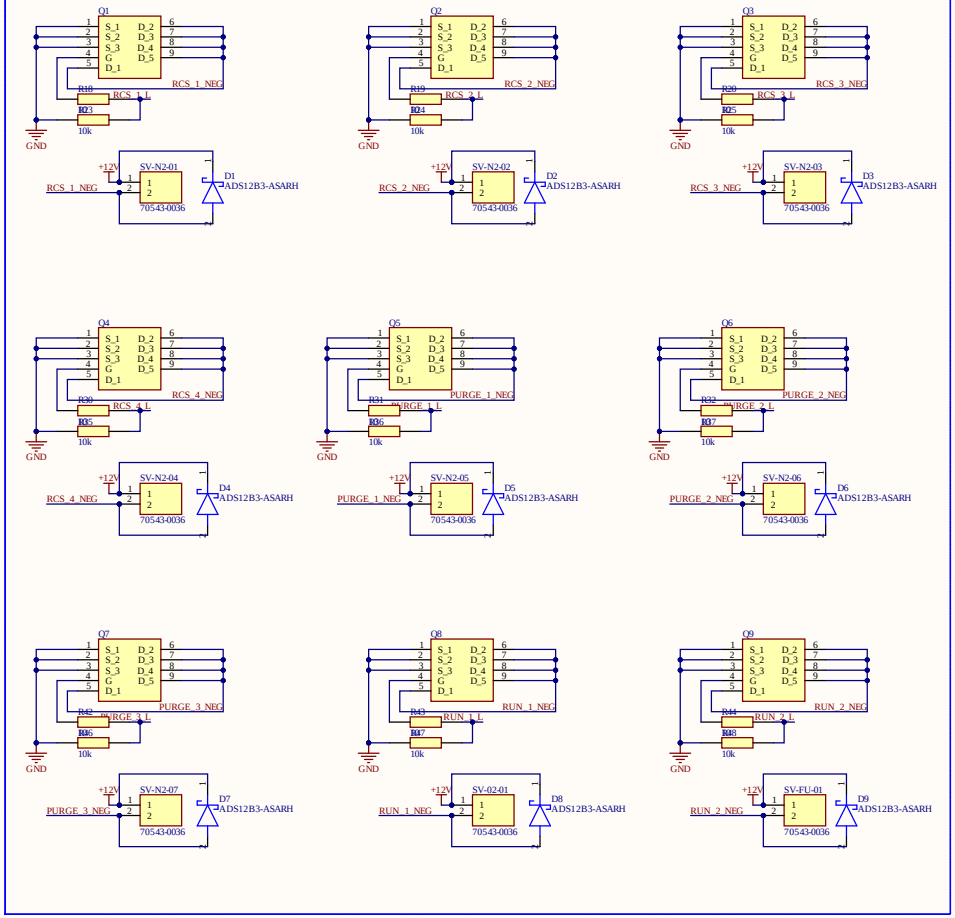
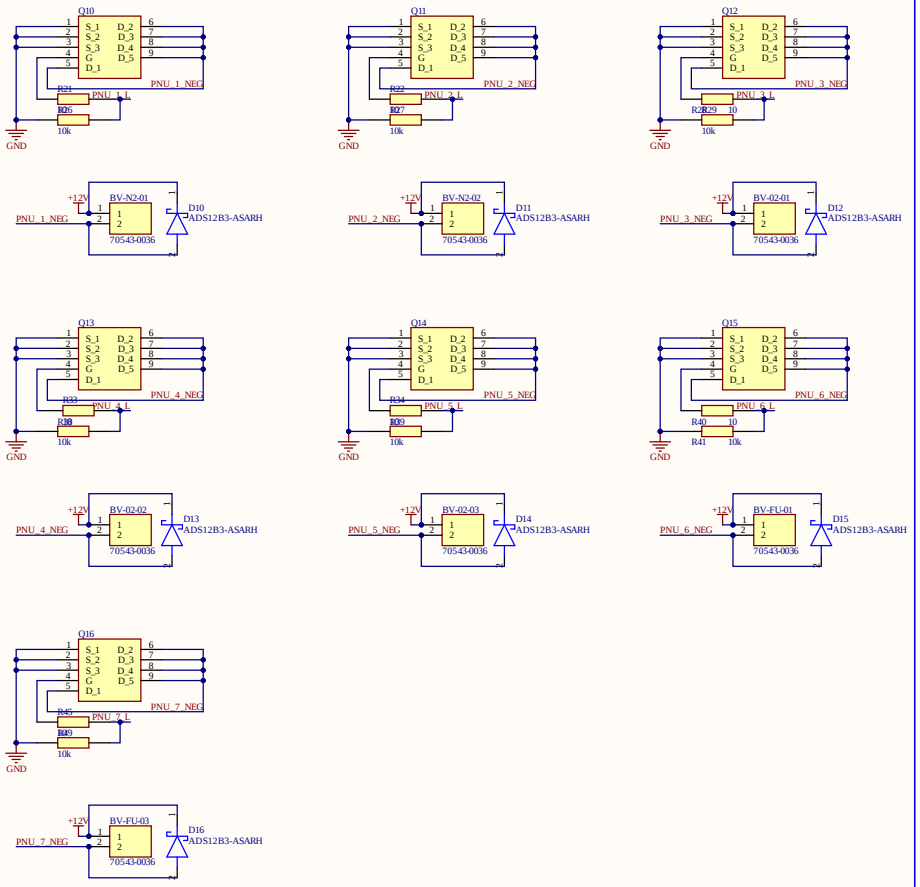

VLV
VLV.SchDoc
SOL_VLV
SOL_VLV
PNU_VLV
PNU_VLV
Termination
Termination
MCU
MCU.SchDoc
Termination
Termination
SPK
SPK
PNU_VLV
SOL_VLV
PNU_VLV
SOL_VLV
CAN1
CAN2
CAN1
CAN2
SPI1
CS_TC
FAULT_TC
SPI1
CS_TC
FAULT_TC
BOOT/RESET
SPI3_G4
UART3_G4
BOOT/RESET
SPI3_G4
UART3_G4
CS_PT
SPI3
ZUCROW
CS_PT
SPI3
ZUCROW
USART2
USART6
USART2
USART6
Spark
SPK.SchDoc
SPK
SPK
PT
PT.SchDoc
SPI1
SPI3
ZUCROW
CS_PT
SPI1
SPI3
ZUCROW
CS_PT
STM32F4
G4.SchDoc
UART3_G4
BOOT/RESET
SPI3_G4
UART3_G4
BOOT/RESET
SPI3_G4
TC
TC.SchDoc
SPI1
SPI3
CS_TC
FAULT_TC
SPI1
SPI3
CS_TC
FAULT_TC
RS1
RS1.SchDoc
USART6
USART6
CAN
CAN.SchDoc
CAN1
CAN2
CAN1
CAN2
RS2
RS2.SchDoc
USART2
USART2
Title
Size
A
Number
Revision
Date:
9/13/2026
Sheet of
File:
C:\Users\..\EC_MainSheet.SchDoc
Drawn By:


STM32H7 - Pins
STM32A
STM32D
STM32B
STM32H747BIT6
MISO to MISO on SPI connection between mcus
Light Blue -> Solenoid and Pnuematic Valves
SPI Lines -> Yellow
UART Lines -> Pink
CAN Lines -> Red
Program STM32 -> Blue
CS TC & PT -> Green
DE/RE -> Light Green
Voltages
VOLTAGE
42819-4213
MCU Signals
BOOT/RESET
SPI1
SPI3_G4
SPI3
CAN1
CAN2
USART2
UART3_G4
USART6
CS_PT
CS_TC
FAULT_TC
SOL_VLV
PNU_VLV
SPK
ZUCROW
STM32H7 - Power
STM32C
128 MB Flash - Connector
Mounting Holes
GPIO
External Communications Protocols
8MHz MEMS - H7
USB1 - SMD
Reset - H7
Boot Mode - H7
LED
Breakouts
Title
MCU.SchDoc
Size
D
Number
2
Revision
3
Date:
9/13/2026
Sheet of
File:
C:\Users\..\MCU.SchDoc
Drawn By:
Javier Aldana

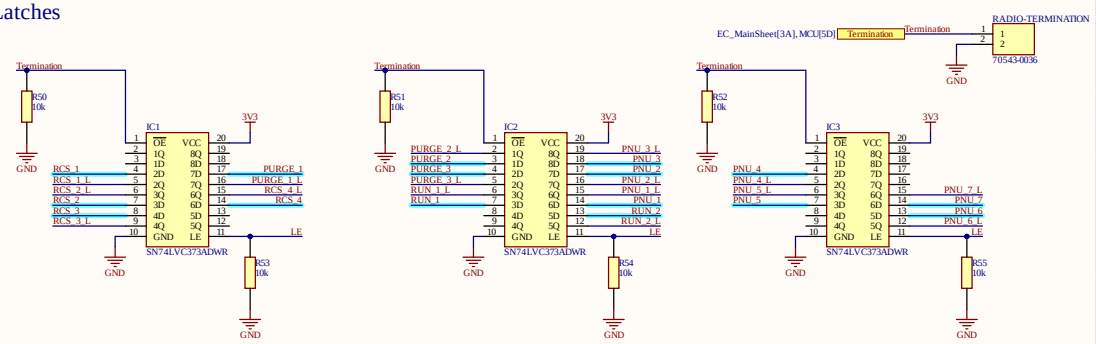
Solenoid Valves



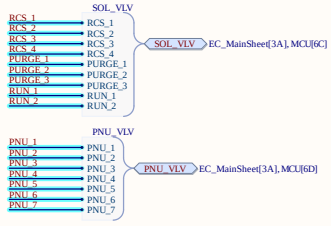
Pneumatic Actuator Valves



Latches



Mcu Signals

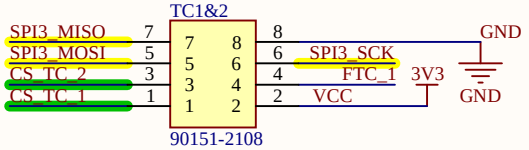


latches: LE pulled low to stop new signal if mcu goes floating
oe grounded to always have current flow
OE pulled low to have current flow, high signal will open circuit

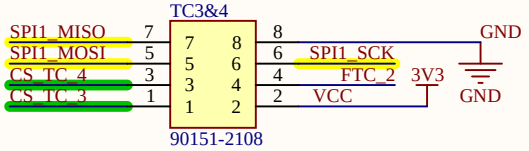
4 RCS Valves, 3 Purge Vavles, 2 Dart Run 7 ball valves (pneumatic)
Each with Flyback Diodes, Lowside Switch

Title	VLV.SchDoc		
Size C	Number 3	Revision 3	
Date: 9/13/2026	Sheet of		
File: C:\Users\..\VLV.SchDoc	Drawn By: Javier Aldana		

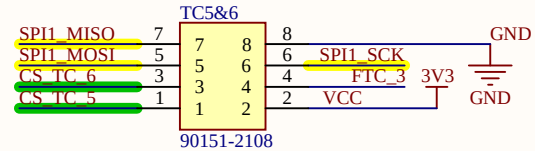
TC1 & TC2



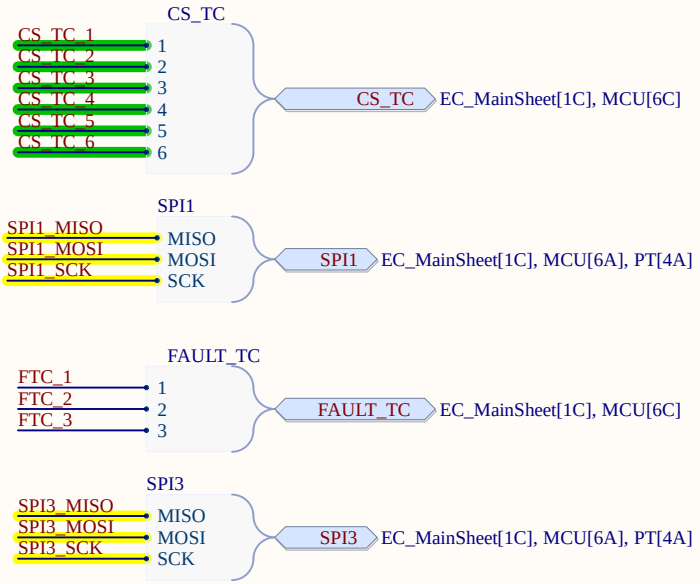
TC3 & TC4



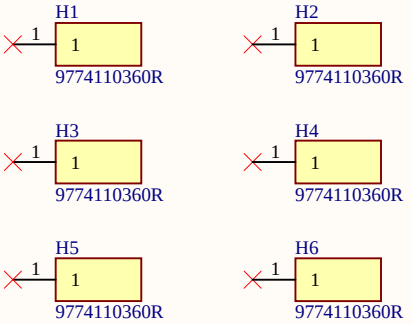
TC5 & TC6



MCU SIGNALS



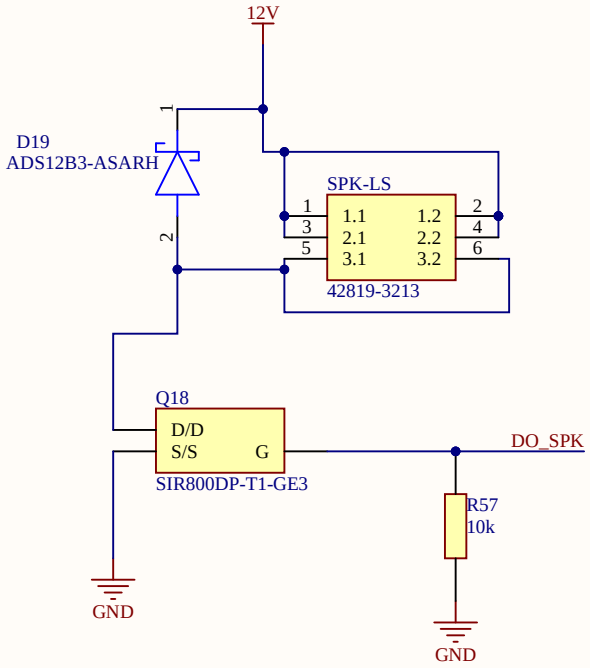
Standoffs



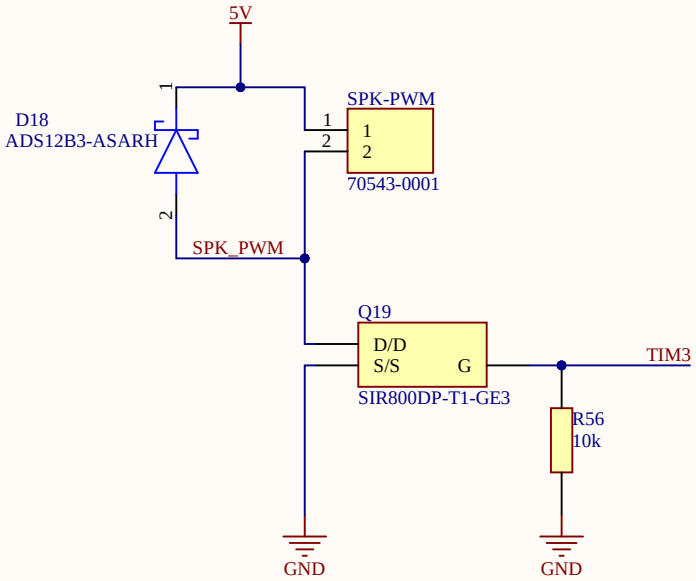
1 fault line per adc,
2 tcs per adc

Title	TC.SchDoc		
Size: A	Number: 4	Revision: 3	
Date: 9/13/2026		Sheet of	
File: C:\Users\..\TC.SchDoc		Drawn By: Javier Aldana	

Lowside Switch - Power



PWM Coil Input



MCU SIGNALS



DO spk turns on Relay that will be normally open
TIM2 turns on FET that drives pwm on spark coil

Title	SPK.SchDoc		
Size A	Number 5	Revision 3	
Date: 9/13/2026		Sheet of	
File: C:\Users\..\SPK.SchDoc		Drawn By: Javier Aldana	

Standoffs

H7 H8 H9 H10 H11
H12 H13 H14 H15 H16
H17 H18 H19 H20

9774110360R

MCU Signals

SPI1: SPI1_MISO, SPI1_MOSI, SPI1_SCK → MISO, MOSI, SCK → SPI1 EC_MainSheet[1B], MCU[6A], TC[3B]

SPI3: SPI3_MISO, SPI3_MOSI, SPI3_SCK → MISO, MOSI, SCK → SPI3 EC_MainSheet[1B], MCU[6A], TC[3B]

CS_PT: CS_PT_3&4, CS_PT_5&6, CS_PT_7&8, CS_PT_9&10, CS_PT_11&12, CS_PT_1&2, PT_Sync → 3, 5, 7, 9, 11, 1, SYNC → CS_PT EC_MainSheet[1B], MCU[6C]

ZUCROW: DO1, DO2, DI1, DI2, DI3, DI4, CS_DAQ, EXTRA → ZUCROW EC_MainSheet[1B], MCU[6D]

PT1 & PT2

PT1&2 — 90151-2108

SPI3_MISO, SPI3_MOSI, SPI3_SCK, CS_PT_1&2, 3V3, 12V, PT_Sync, GND

PT3 & PT4

PT3&4 — 90151-2108

SPI3_MISO, SPI3_MOSI, SPI3_SCK, CS_PT_3&4, 3V3, 12V, PT_Sync, GND

PT5 & PT6

PT5&6 — 90151-2108

SPI3_MISO, SPI3_MOSI, SPI3_SCK, CS_PT_5&6, 3V3, 12V, PT_Sync, GND

PT7 & PT8

PT7&8 — 90151-2108

SPI3_MISO, SPI3_MOSI, SPI3_SCK, CS_PT_7&8, 3V3, 12V, PT_Sync, GND

PT9 & PT10

PT9&10 — 90151-2108

SPI1_MISO, SPI1_MOSI, SPI1_SCK, CS_PT_9&10, 3V3, 12V, PT_Sync, GND

PT11 & PT12

PT11&12 — 90151-2108

SPI1_MISO, SPI1_MOSI, SPI1_SCK, CS_PT_11&12, 3V3, 12V, PT_Sync, GND

SYNC PU

3V3 → R58 10k → PT_Sync

3V3 → DNP 10k → DO1

PT13 & PT14 / Zucrow

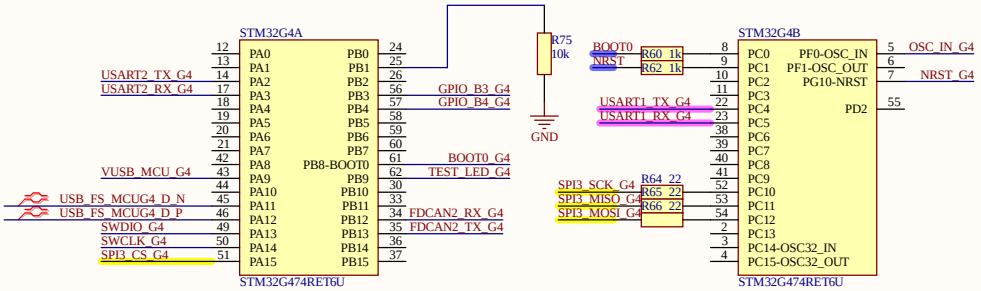
PT13&14/Zucrow — 90151-2114

SPI1_MISO, SPI1_MOSI, SPI1_SCK, DO2, DI1, DI3, CS_DAQ, 3V3, 12V, DO1, DI2, DI4, EXTRA, GND

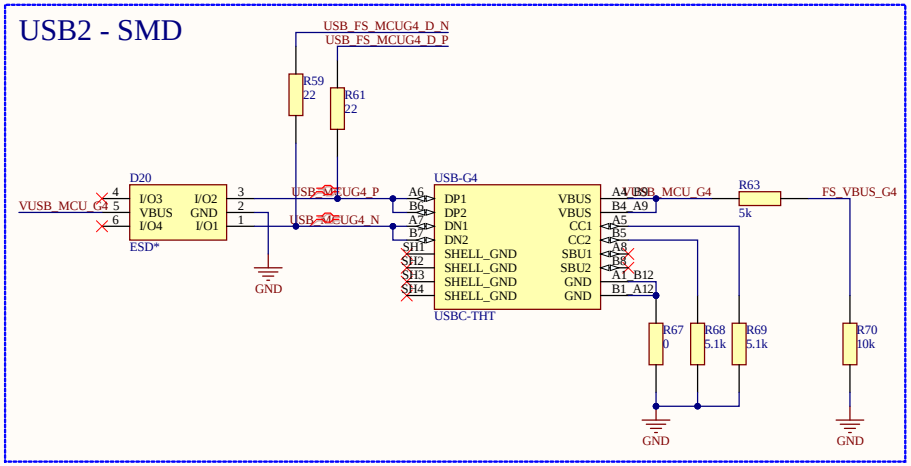
Title	PT.SchDoc		
Size: A	Number		
Date:	9/13/2026	Sheet	of
File:	C:\Users\..\PT.SchDoc	Drawn By:	Javier Aldana

have on pt be able to do : 4 DI, 2 DO, 2AO (from EC)

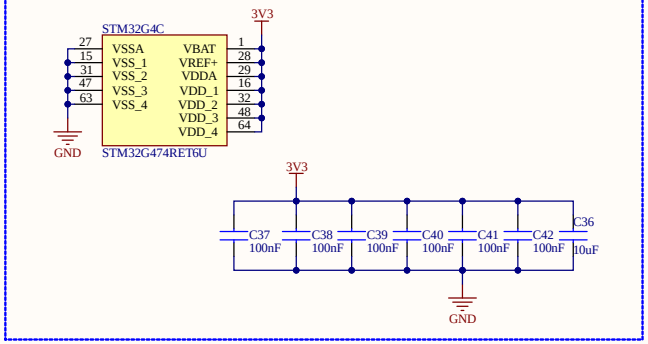
STM32G474 - Pins



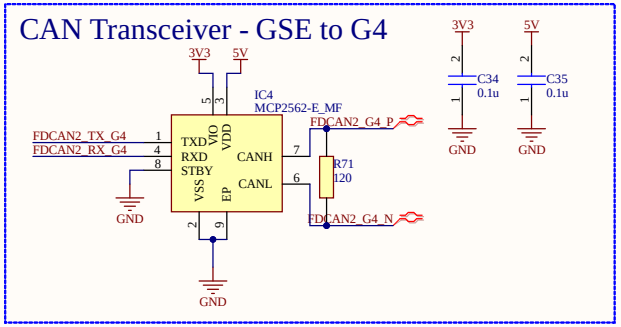
USB2 - SMD



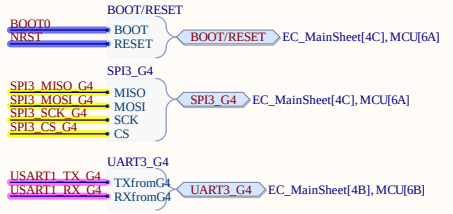
STM32G474 - Power



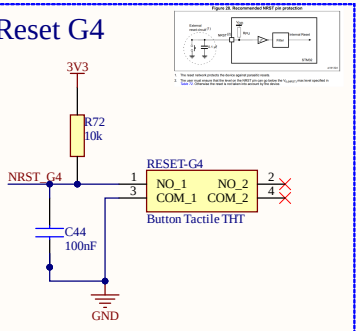
CAN Transceiver - GSE to G4



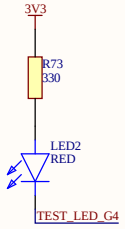
MCU Signals



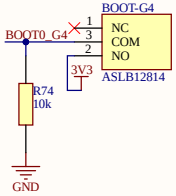
Reset G4



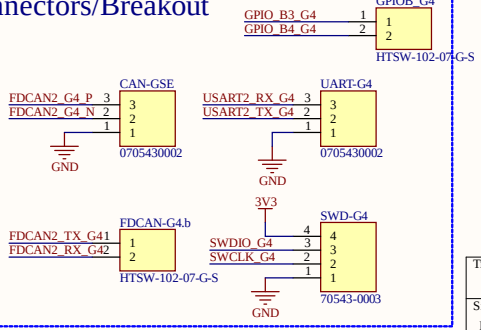
LED G4



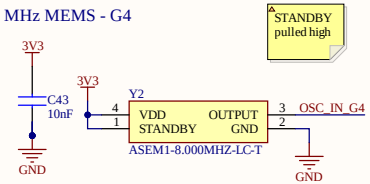
BOOT Mode - G4



Connectors/Breakout

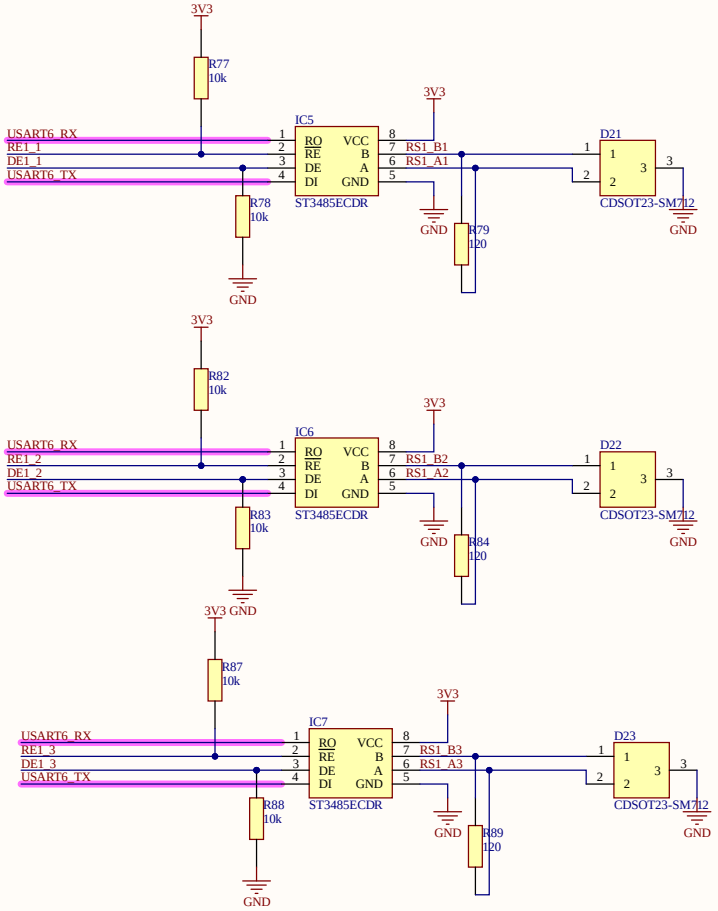


8 MHz MEMS - G4

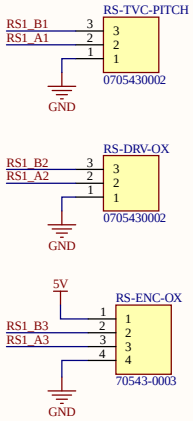


Title	G4.SchDoc		
Size: B	Number: 6	Revision: 3	
Date: 9/13/2026		Sheet of	
File: C:\Users\..\G4.SchDoc		Drawn By: Javier Aldana	

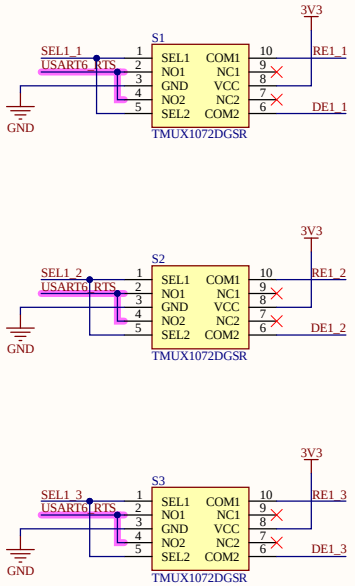
RS485 Transceiver - Driver, Encoder, TVC



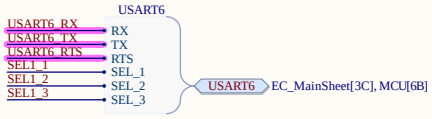
RS485 Connectors



MUXes



MCU Signals



Javier Aldana

Title: RS1.SchDoc		
Size: B	Number: 7	Revision: 3
Date: 9/13/2026		Sheet of
File: C:\Users\..\RS1.SchDoc		Drawn By:

RS485 Transceiver - Driver2, Encoder2, TVC2

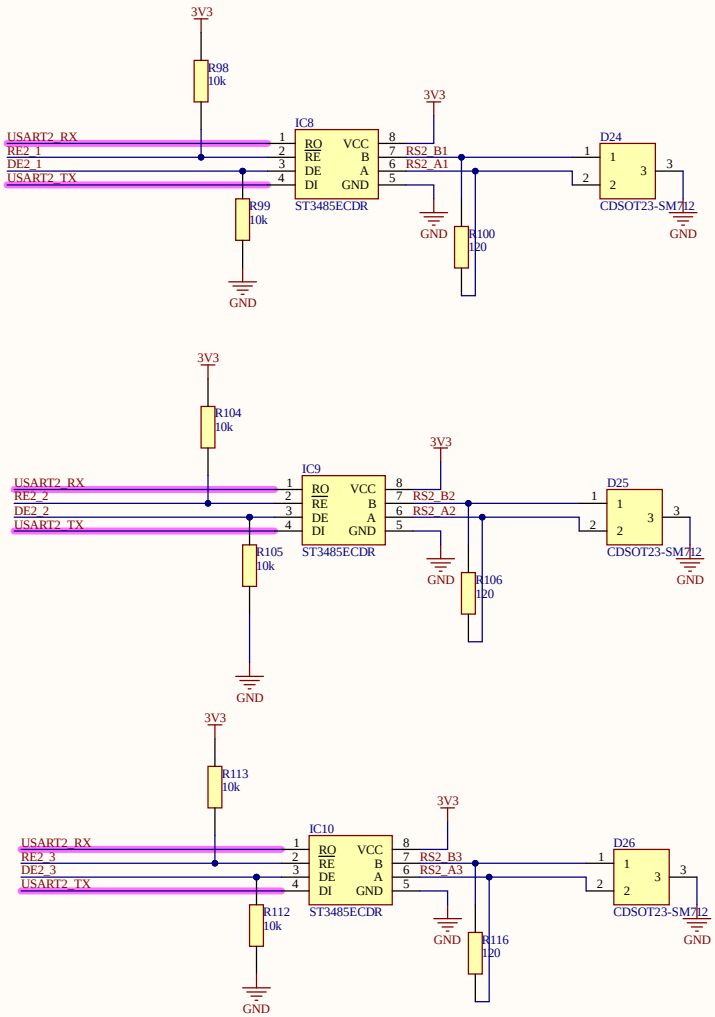
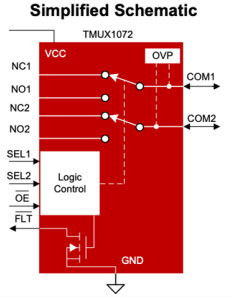


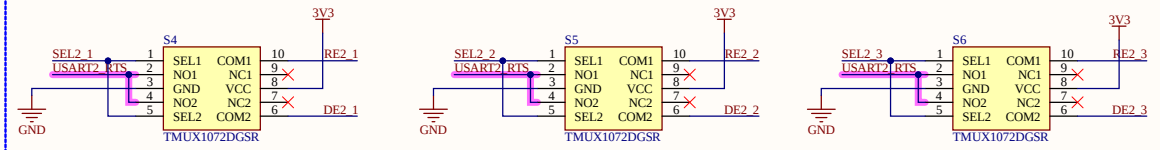
Table 1. Function Table

$\overline{OE}$	SEL1	SEL2	COM1 Connection	COM2 Connection
H	X	X	High-Z	High-Z
L	L	L	COM1 to NC1	COM2 to NC2
L	L	H	COM1 to NC1	COM2 to NO2
L	H	L	COM1 to NO1	COM2 to NC2
L	H	H	COM1 to NO1	COM2 to NO2

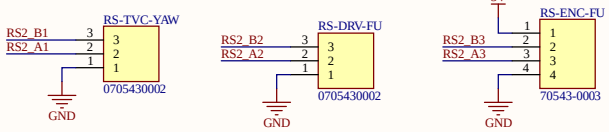
Simplified Schematic



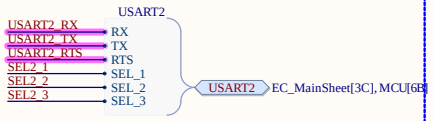
MUXes



RS485 Connectors

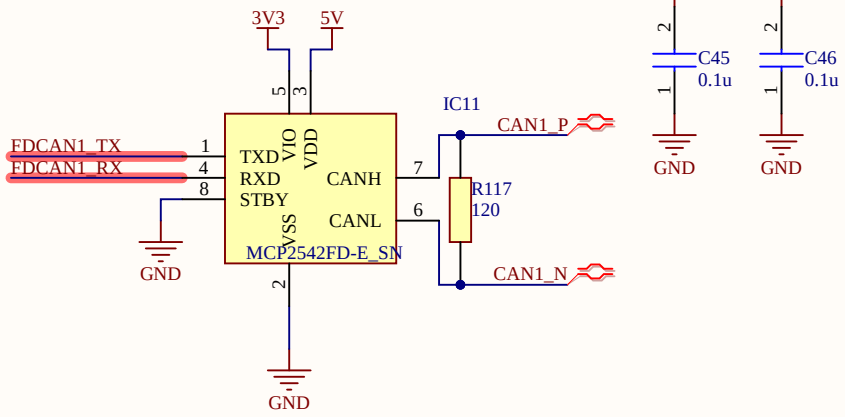


MCU Signals

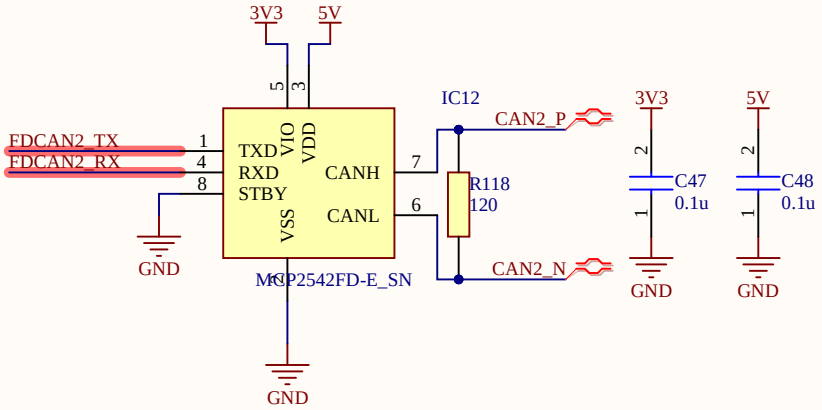


Title	RS2.SchDoc		
Size: B	Number: 8	Revision: 3	
Date: 9/13/2026		Sheet of	
File: C:\Users\..\RS2.SchDoc		Drawn By: Javier Aldana	

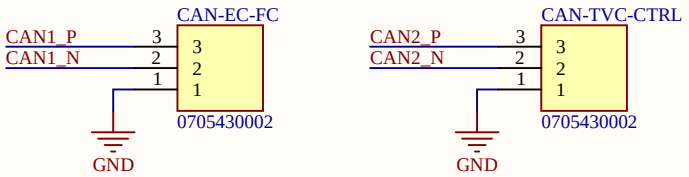
CAN Transceiver - CAN1



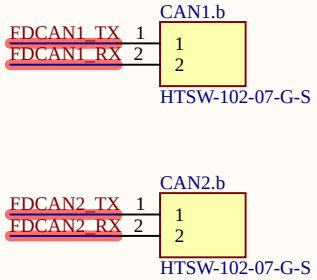
CAN Transceiver - CAN2



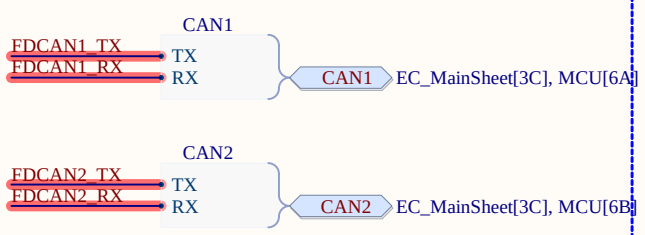
Connectors



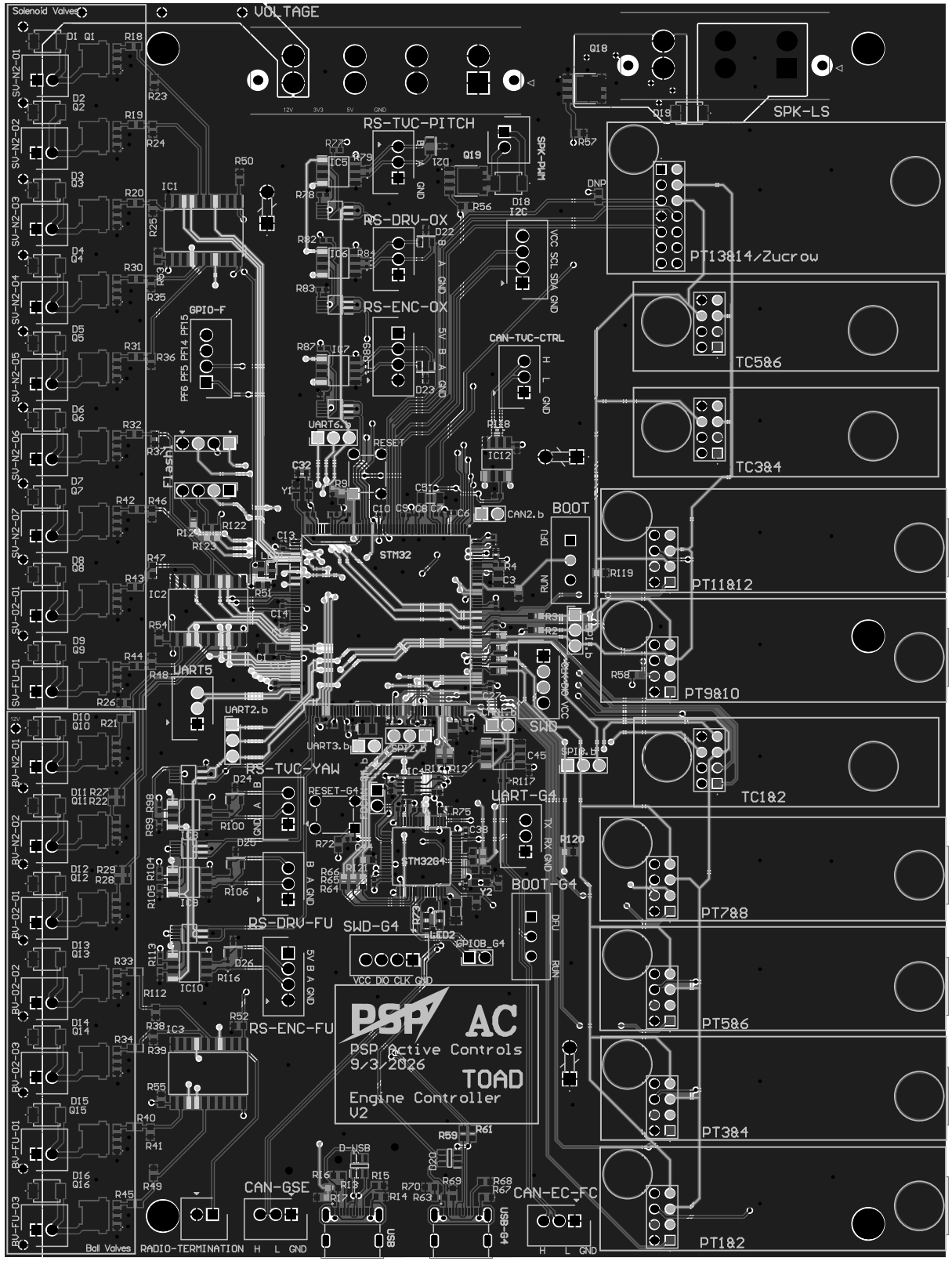
Breakouts



MCU Signals



Title	CAN.SchDoc		
Size: A	Number: 9	Revision: 3	
Date: 9/13/2026		Sheet of	
File: C:\Users\..\CAN.SchDoc		Drawn By: Javier Aldana	


Solenoid Valves
VOLTAGE
12V 3V3 5V GND
SPK-LS
RS-TVC-PITCH
RS-DRV-OX
RS-ENC-OX
SPK-PWM
I2C
VCC SCL SDA GND
CAN-TVC-CTRL
H L GND
GPIO-F
PF6 PF5 PF14 PF15
Flash
UART6.b
RESET
STM32
BOOT
DFU
RUN
CAN2.b
SWD
SPI2.b
SPI6.b
UART5
UART2.b
UART3.b
RS-TVC-YAW
RESET-G4
UART-G4
TX RX GND
STM32G4
BOOT-G4
RS-DRV-FU
SWD-G4
SPIOB_G4
VCC DIO CLK GND
RS-ENC-FU
PSP AC
PSP Active Controls
9/3/2026
TOAD
Engine Controller
V2
D-USB
CAN-GSE
CAN-EC-FC
USB
USB-G4
RADIO-TERMINATION H L GND
H L GND
Ball Valves
PT13&14/Zucrow
TC5&6
TC3&4
PT11&12
PT9&10
TC1&2
PT7&8
PT5&6
PT3&4
PT1&2
SV-N2-01
SV-N2-02
SV-N2-03
SV-N2-04
SV-N2-05
SV-N2-06
SV-N2-07
SV-O2-01
SV-FU-01
BV-N2-01
BV-N2-02
BV-O2-01
BV-O2-02
BV-O2-03
BV-FU-01
BV-FU-02
BV-FU-03

Item	Name	Description	Footprint	Quantity	Manufacturer	Manufacturer Number	Supplier	Supplier Part Number	Supplier Unit Price	Supplier Extended Price
	105-00108-xx	Switch	SOT, SOT-6x		Not managed			2432-105-00108-0.1	https://valexa.sqlite.cl/column/105-00108-xx/ https://valexa.sqlite.cl/table/105_SPST-Momentary-Switches/ 19.8.3.pdf?sanitized-u0bf1a87a1ff5c7d1f1f8f1e1f1f1f